

Probability – 3

Random walks, Markov chains and limit theorems

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Theoretical minimum.

Def. A *Markov chain* on a finite or countable set is a sequence with $\Pr(X_{n+1} = y \mid X_n = x, \dots) = p_{xy}$. A state is *recurrent* if the walk returns to it with probability one, otherwise *transient*. A distribution π is *stationary* if $\pi P = \pi$.

Theorem 5 (gambler’s ruin and recurrence). For the simple symmetric walk on $\mathbb{Z}$ started at k between 0 and N the probability of reaching N before 0 is k/N , and the expected duration is $k(N - k)$. The simple symmetric walk is recurrent on $\mathbb{Z}$ and on $\mathbb{Z}^2$ and transient on $\mathbb{Z}^3$; the proof compares $\sum_n \Pr(X_{2n} = 0)$ with $\sum n^{-d/2}$.

Theorem 6 (finite chains). An irreducible finite chain has a unique stationary distribution, and if it is also aperiodic, then $\Pr(X_n = y) \rightarrow \pi_y$ for every start. The expected return time to x equals $1/\pi_x$. The standard proof is coupling two copies of the chain.

Theorem 7 (branching processes). If each individual independently leaves offspring with generating function $f(s) = \sum p_k s^k$, then the extinction probability is the smallest root of $f(s) = s$ in $[0, 1]$; it equals one exactly when the mean offspring number is at most one.

Fact 3 (limit theorems and when they are needed). The law of large numbers and the Borel–Cantelli lemmas settle almost sure statements; the central limit theorem is rarely needed in an olympiad, but concentration is: in high dimension two independent uniform unit vectors are nearly orthogonal, and this is exactly the statement that a sum of n independent terms of size $n^{-1/2}$ concentrates.

1. Prove Theorem 5 in full, including the recurrence of the walk on $\mathbb{Z}^2$ and its transience on $\mathbb{Z}^3$, and prove the formula $1/\pi_x$ for the expected return time of a finite chain.
2. (ISTCiM 2022) A bacterium splits every second into two perfect copies with probability p , and otherwise disintegrates; all copies behave independently. Find the probability that a colony started by one bacterium never dies out.
3. (ISTCiM 2024) A flea starts at $(0, 0)$ on the grid $\mathbb{Z}^2$ and first jumps to $(1, 0)$; afterwards at each vertex it jumps to one of the three neighbours other than the one it came from, each with probability $\frac{1}{3}$, independently. Find the expected square of its distance from the origin after n jumps.
4. (S.T. Yau CSMC 2016) For the random walk on the complete infinite binary tree started at the root, which moves to each neighbour with equal probability, let X_n be the level at time n . Prove that $\mathbb{E} X_n \leq \frac{n}{3} + \frac{4}{3}$.
5. (VJIMC 2008) A player starts with capital $x_0 \in (0, C)$, where $C > 2$. If the current capital is x , put $s(x) = x$ when $x < 1$, $s(x) = C - x$ when $C - x < 1$, and $s(x) = 1$ otherwise; a fair coin then increases or decreases the capital by $s(x)$. Find the probability that the player reaches C in finitely many turns.